Venkata Lakshmi Parimala Pasupuleti

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Virginia, USA (Remote)

PROFESSIONAL SUMMARY

AI Agent specialist with 8+ years serving as primary technical consultant for enterprise customers and hands-on experience designing and building production agentic AI systems. I've built LeaseI -- an 8-agent LangGraph pipeline with multi-system orchestration, conversational AI workflows, and human-in-the-loop escalation (93.6% precision on 1,240 test cases) -- and AgentProbe, an open-source AI evaluation toolkit used to measure agent performance in production. At Entrata, I was the go-to technical resource for new enterprise customers pre-go-live: translating multi-system business logic into platform solutions, leading technical discovery with operations teams, building the business case for implementation scope, and owning the technical narrative end-to-end through deployment. Equally comfortable whiteboarding agent architectures with an engineering team or explaining AI Agent value to executive stakeholders. Strong in Python, LangGraph, LLM orchestration patterns, and API integration across enterprise systems. AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

AI Agent Design & Building | Agentic AI & Orchestration Patterns | Enterprise Technical Consulting | Workshop Facilitation & Use Case Design | API Discovery & Integration Patterns | LLM-Based Architecture | Technical Discovery & Proof-of-Value | Conversational AI & NLP Systems

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building agentic AI pipelines and integrating data systems; translating recurring customer implementation patterns into reusable product components; serving as technical resource bridging AI capabilities to business outcomes.

Magizhs Tech Solution
May 2025 -- Aug 2025

AI/ML Engineer Intern (Forward Deployed)

- **Served as primary AI Agent technical consultant end-to-end:** embedded with operations teams, led technical discovery workshops to map multi-system business logic into AI Agent use cases, built a LangGraph agentic pipeline (parse → plan → retrieve → validate → routing), integrated directly with client enterprise systems via APIs, and owned the technical narrative through deployment and rollout.
- Built event-driven validation guardrails enforcing business rules before output delivery; implemented RAG pipeline with cited compliance decisions; translated implementation learnings into reusable product components. Served as the technical bridge between AI capabilities and client proof-of-value.

George Mason University
Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Designed and facilitated technical workshops for 600+ students across multiple sections -- translating complex AI and data concepts into accessible, outcome-focused learning. Delivered one-on-one technical coaching; built multi-source prediction pipelines (Python, R, $R^2 = 0.977$); fine-tuned a custom classification model (HuggingFace + PyTorch).

Entrata (Enterprise PropTech SaaS -- 100K+ monthly active users)
Jan 2017 -- Aug 2024

Senior Software Engineer + Enterprise Technical Consultant -- Lehi, UT (Remote)

- **Owned the technical narrative for new enterprise customer onboardings end-to-end:** embedded with operations teams, conducted technical discovery to surface integration requirements and API feasibility, translated multi-system business logic into platform solutions, wrote per-team technical specs, built the business case and ROI model for implementation scope, and coordinated all cross-system dependencies before go-live. Functioned as the pre-sales technical owner for every new account. Zero compliance failures across every deployment.
- **Go-to internal expert on integration patterns and enterprise API workflows:** built and maintained the integration layer connecting property management operations across 5+ enterprise systems -- REST APIs, webhook endpoints, event-driven async pipelines (RabbitMQ), CRM connectors, scheduling workflows, and third-party data source integrations for 100K+ monthly active users at 99.5% uptime. Elevated team fluency through 300+ code reviews and internal debugging tooling that cut debug time 40%.
- Synthesized patterns from enterprise customer deployments, fed structured feedback to product and engineering, and turned recurring implementation learnings into reusable platform components -- the same pre-sale to deployment feedback loop this role requires.

PROJECTS

LeaseI: Production Agentic AI Platform -- 8-Agent LangGraph Pipeline
LangGraph -- FastAPI -- RAG -- Conversational AI -- HITL

Designed and built an end-to-end agentic AI system using LLM-based architecture: 8-agent LangGraph pipeline with durable state management and conditional routing (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Human Escalation). Multi-system orchestration across structured databases, document stores, configuration systems, and regulatory rule engines. Conversational AI interface for tenant self-service; human-in-the-loop escalation when agent confidence is too low; guardrails blocking unsupported outputs. Results: 93.6% precision / 87.9% F1 on 1,240 test cases; 100% citation correctness. Built with full product ownership, no PM, no roadmap.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: Open-Source AI Agent Evaluation & Observability Toolkit

FastAPI -- LangGraph -- CI/CD -- Multi-Provider LLM

Production evaluation toolkit measuring whether deployed AI agents achieve their operational objectives: 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate with golden datasets, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), observability dashboard. Used to evaluate LeaseI across 1,240+ test cases. Built to close the agent performance feedback loop.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: Conversational AI Integration Layer for Clinical Systems

Flask -- LLMs -- BERT -- REST API -- NLP

Built a conversational AI platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings -- direct API integration with clinical records systems, HIPAA-compliant RBAC, and NLP-driven use case discovery across multi-system data stores. ~60% reduction in manual retrieval time. Owned the technical narrative through customer adoption.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

AI Agents & Orchestration: LangChain, LangGraph, Agentic AI, Multi-Agent Orchestration, Orchestration Patterns, LLM-Based Architecture, Agent Frameworks, Conversational AI, HITL, RAG Pipelines, Prompt Engineering, Function Calling, AI Guardrails, Observability, FastAPI, MCP, OpenAI API, Anthropic API

Integration & APIs: REST APIs, API Discovery, Integration Patterns, Webhook Endpoints, Enterprise System Integration, CRM Connectors, Event-Driven Architecture, RabbitMQ, OAuth, OpenAPI, WebSockets, Third-Party Integrations

Languages & Frameworks: Python, TypeScript, JavaScript, React.js, Node.js, SQL, FastAPI, Flask, Django, Java, Spring Boot

Cloud & Infra: AWS (S3, EC2, Lambda, Glue), Docker, Kubernetes, CI/CD (GitHub Actions), PostgreSQL, Redis, Linux